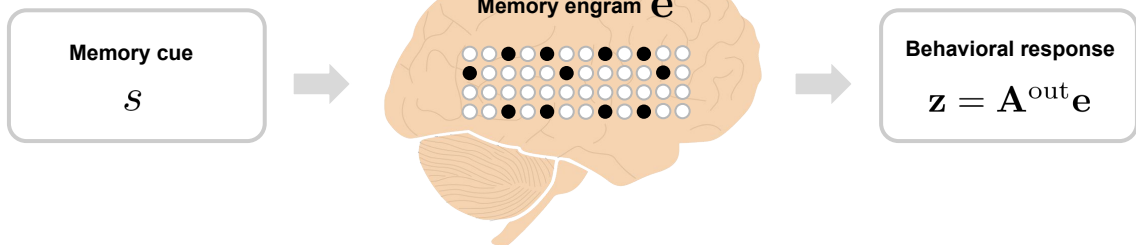


A

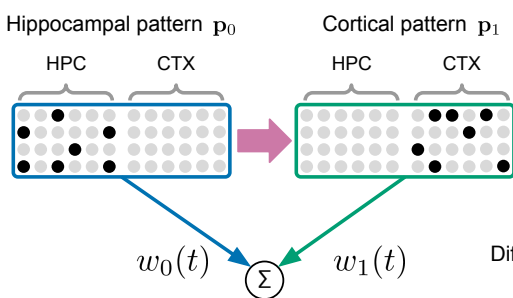
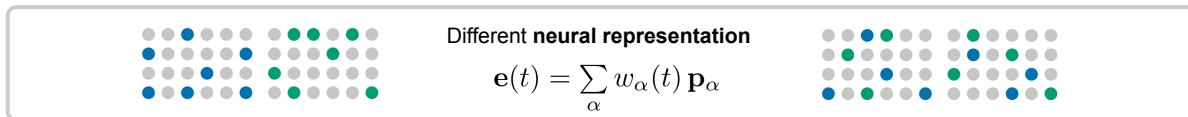
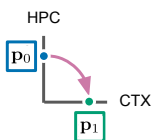
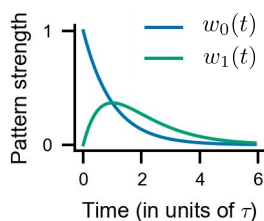
Systems memory consolidation:
Reactivation changes the engram

$$\frac{d}{dt} e = -\tau^{-1} e + \mathbf{A} e$$

↑ change = ↑ forgetting + ↑ consolidation

B

$$\mathbf{p}_0 = \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} = \begin{pmatrix} \text{blue} \\ 0 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} \text{HPC} & \text{CTX} \\ 0 & 0 \\ \text{pink} & 0 \end{pmatrix}, \quad \mathbf{p}_1 = \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} = \begin{pmatrix} 0 \\ \text{green} \end{pmatrix}$$

C**D****E****F**

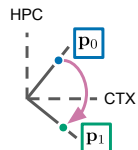
Same pattern strength dynamics

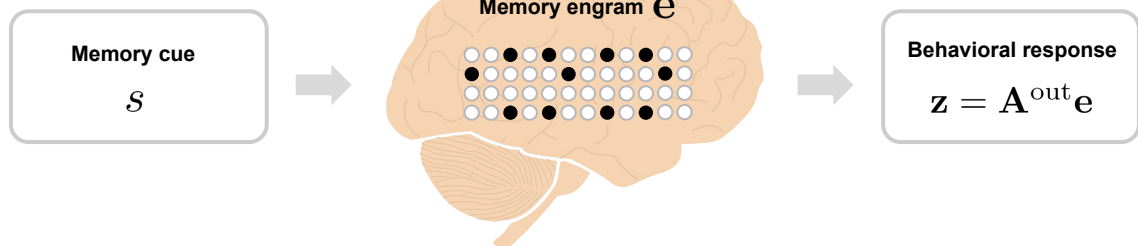
$$\frac{d}{dt} \mathbf{w} = -\tilde{\tau}^{-1} \mathbf{w} + \tilde{\mathbf{A}} \mathbf{w}$$

$$\tilde{\mathbf{p}}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\tilde{\mathbf{p}}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\tilde{\mathbf{A}} = \begin{pmatrix} 0 & 0 \\ \text{pink} & 0 \end{pmatrix}$$

I

A

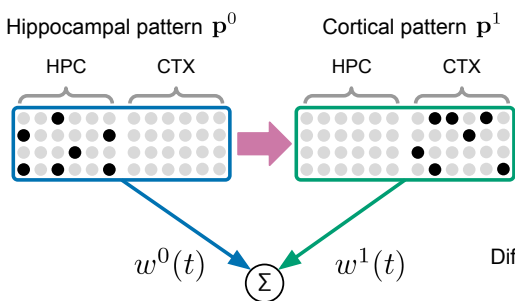
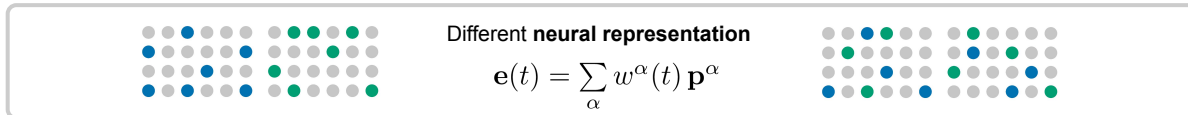
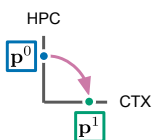
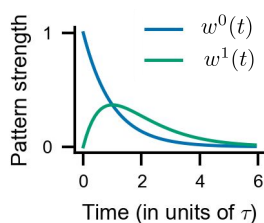
Systems memory consolidation:
Reactivation changes the engram

$$\frac{d}{dt} e = -\tau^{-1} e + \Lambda e$$

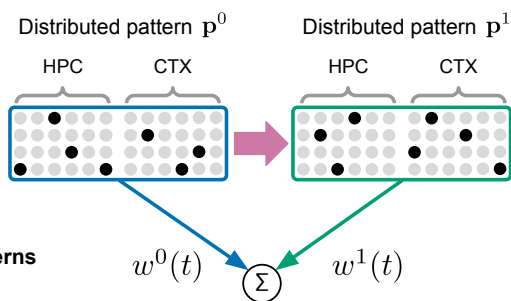
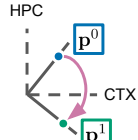
↑ change = ↑ forgetting + ↑ consolidation

B

$$p^0 = \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} \quad A = \begin{pmatrix} \text{HPC} & \text{CTX} \\ \text{HPC} & \text{CTX} \end{pmatrix} \quad p^1 = \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix}$$

C**D****E****F****G**

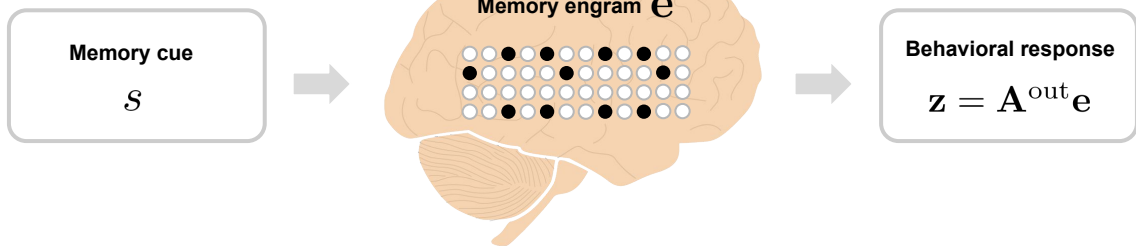
$$p^0 = \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} \quad A = \begin{pmatrix} \text{HPC} & \text{CTX} \\ \text{HPC} & \text{CTX} \end{pmatrix} \quad p^1 = \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix}$$

H**I**

$$\frac{d}{dt} \mathbf{w} = -\tilde{\tau}^{-1} \mathbf{w} + \tilde{A} \mathbf{w}$$

$$\tilde{p}^0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \tilde{p}^1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\tilde{A} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

A

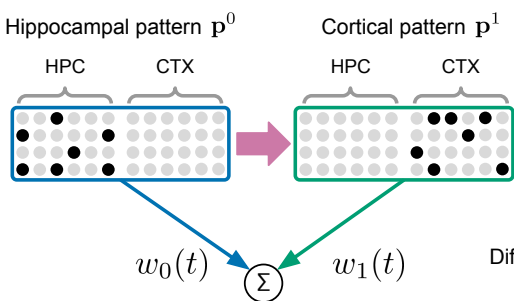
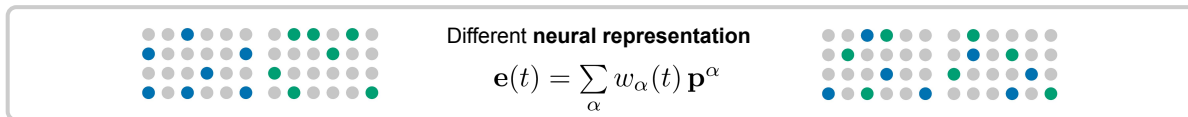
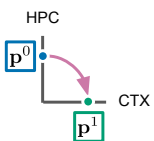
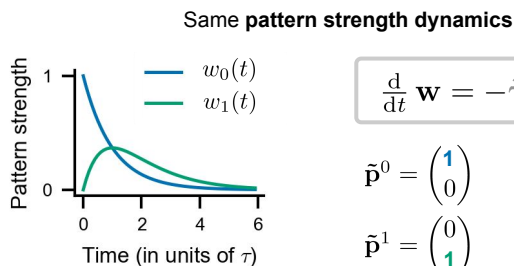
Systems memory consolidation:
Reactivation changes the engram

$$\frac{d}{dt} e = -\tau^{-1} e + \Lambda e$$

$\uparrow$ change = $\uparrow$ forgetting + $\uparrow$ consolidation

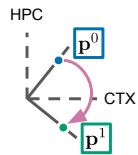
B

$$p^0 = \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} \quad A = \begin{pmatrix} \text{HPC} & \text{CTX} \\ \text{HPC} & \text{CTX} \end{pmatrix} \quad p^1 = \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix}$$

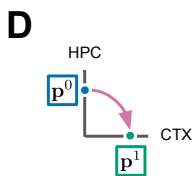
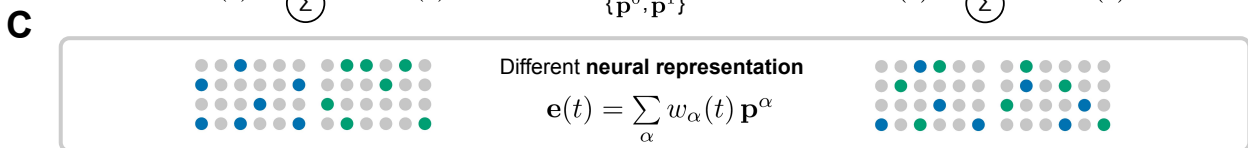
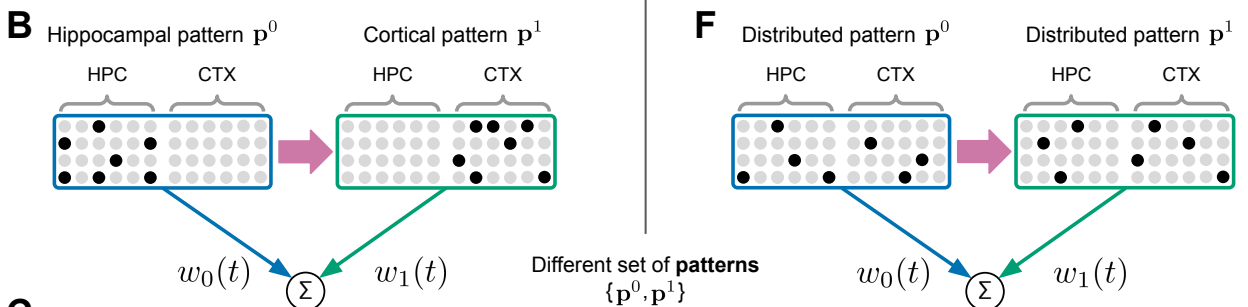
C**D****E****F**

$$\frac{d}{dt} \mathbf{w} = -\tilde{\tau}^{-1} \mathbf{w} + \tilde{\Lambda} \mathbf{w}$$

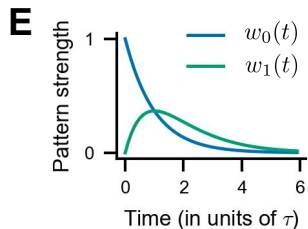
$$\tilde{p}^0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \tilde{\Lambda} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad \tilde{p}^1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

I

$$\mathbf{p}^0 = \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \mathbf{A} = \begin{pmatrix} \text{HPC} & \text{CTX} \\ \text{HPC} & \text{CTX} \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad \mathbf{p}^1 = \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

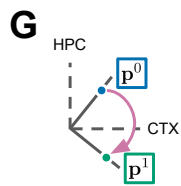


E Same pattern strength dynamics



$$\frac{d}{dt} \mathbf{w} = -\tilde{\tau}^{-1} \mathbf{w} + \tilde{\mathbf{A}} \mathbf{w}$$

$$\tilde{\mathbf{p}}^0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \tilde{\mathbf{A}} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \quad \tilde{\mathbf{p}}^1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

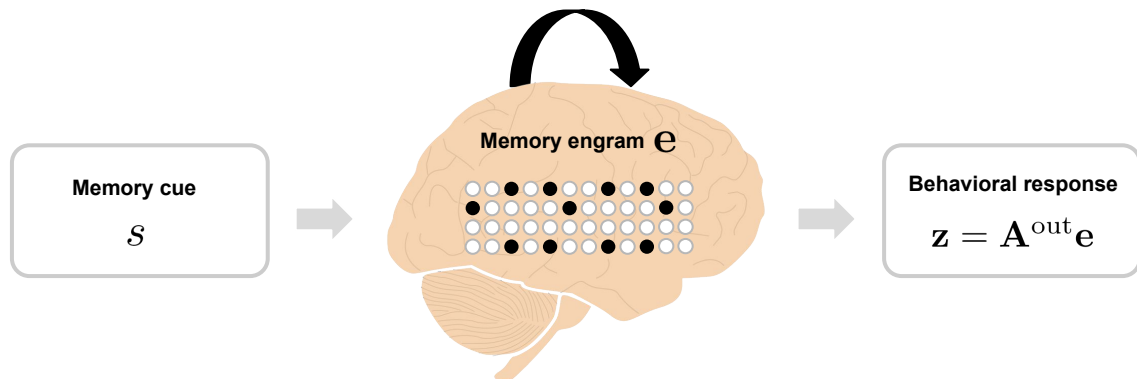
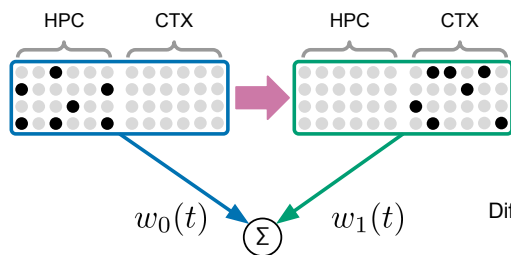
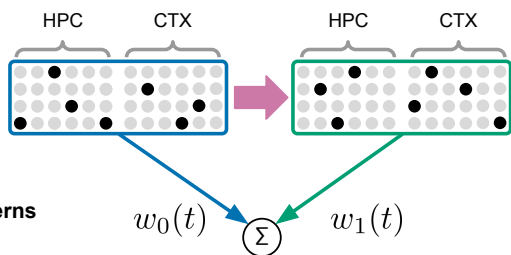


A**Systems memory consolidation:**

Reactivation changes the engram

$$\frac{d}{dt} \mathbf{e} = -\tau^{-1} \mathbf{e} + \mathbf{A} \mathbf{e}$$

↑ ↑ ↑
 change = forgetting + consolidation

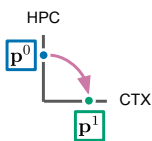
**B**Hippocampal pattern $\mathbf{p}^0$ Cortical pattern $\mathbf{p}^1$ **F**Distributed pattern $\mathbf{p}^0$ Distributed pattern $\mathbf{p}^1$ **C**

Different neural representation

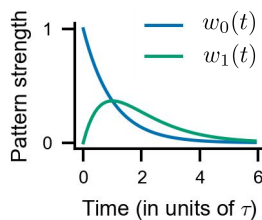
$$\mathbf{e}(t) = \sum_{\alpha} w_{\alpha}(t) \mathbf{p}^{\alpha}$$

**D**

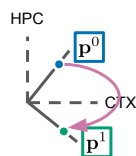
$$\mathbf{A} = \begin{pmatrix} \text{HPC} & \text{CTX} \\ \begin{pmatrix} 0 & 0 \\ \text{pink} & 0 \end{pmatrix} & \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} \end{pmatrix}$$

**E**

Same pattern strength dynamics

**G**

$$\mathbf{A} = \begin{pmatrix} \text{HPC} & \text{CTX} \\ \begin{pmatrix} \text{pink} & \text{pink} \\ \text{pink} & \text{pink} \end{pmatrix} & \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} \end{pmatrix}$$

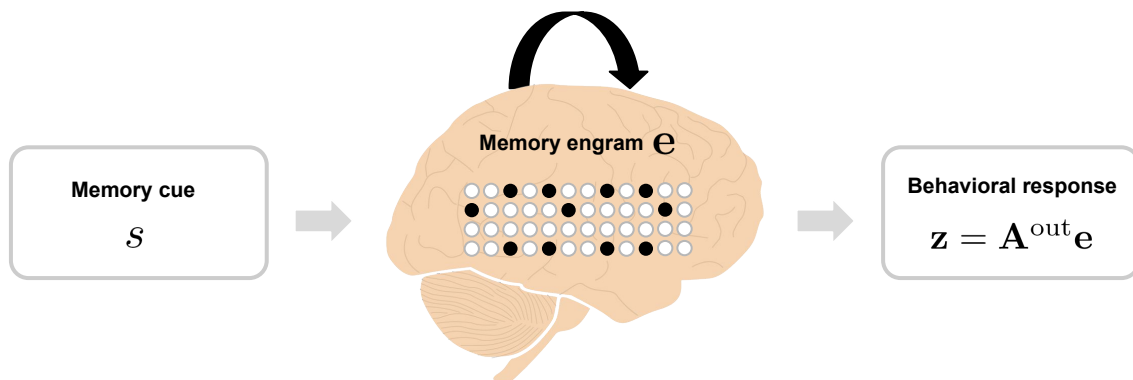
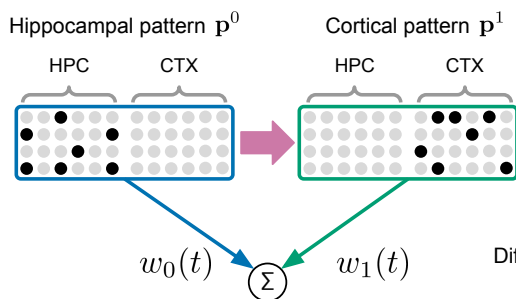
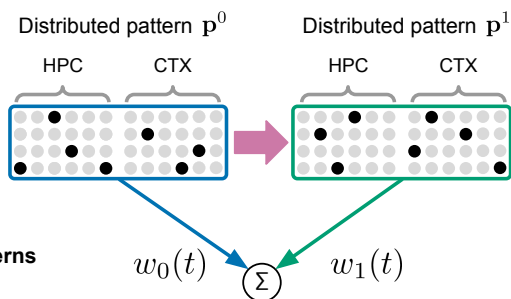
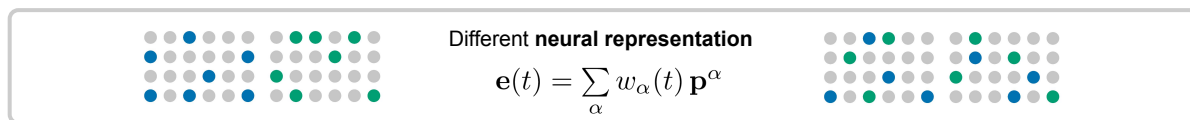


A**Systems consolidation:**

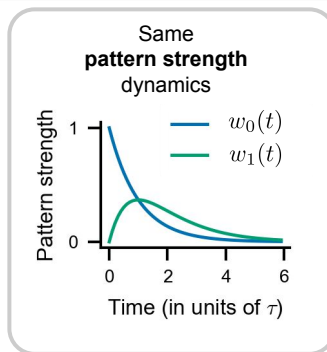
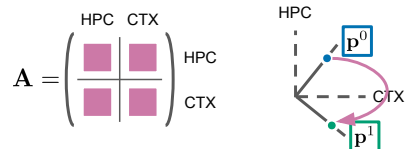
Reactivation changes
the engram

$$\frac{d}{dt} \mathbf{e} = -\tau^{-1} \mathbf{e} + \mathbf{A} \mathbf{e}$$

$\uparrow$ $\uparrow$ $\uparrow$
change = **forgetting** + **consolidation**

**B****F****C****D**

$$\mathbf{A} = \begin{pmatrix} \text{HPC} & \text{CTX} \\ \begin{pmatrix} 0 & 0 \\ \text{pink} & 0 \end{pmatrix} & \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} \end{pmatrix}$$

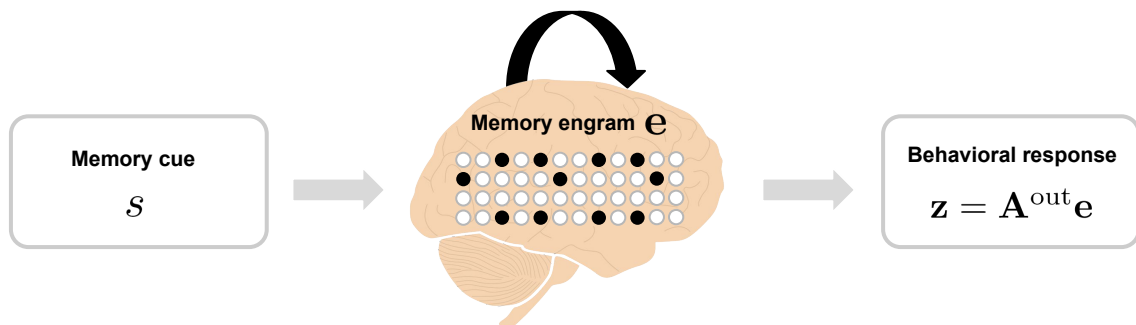
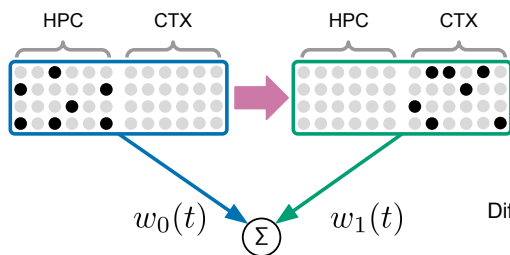
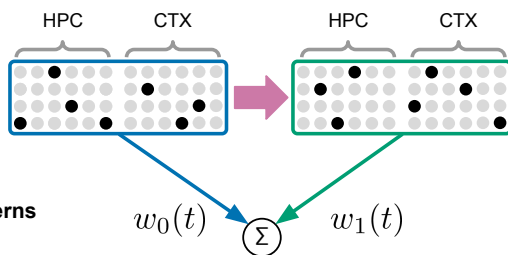
E**G**

A**Systems memory consolidation:**

Reactivation changes the engram

$$\frac{d}{dt} \mathbf{e} = -\tau^{-1} \mathbf{e} + \mathbf{A} \mathbf{e}$$

$\uparrow$
 change = $\uparrow$ forgetting + $\uparrow$ consolidation

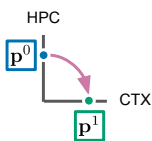
**B**Hippocampal pattern $\mathbf{p}^0$ Cortical pattern $\mathbf{p}^1$ **F**Distributed pattern $\mathbf{p}^0$ Distributed pattern $\mathbf{p}^1$ **C**

Different neural representation

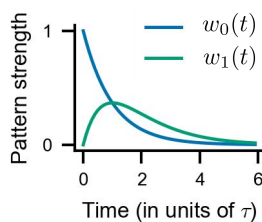
$$\mathbf{e}(t) = \sum_{\alpha} w_{\alpha}(t) \mathbf{p}^{\alpha}$$

**D**

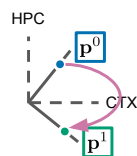
$$\mathbf{A} = \begin{pmatrix} \text{HPC} & \text{CTX} \\ \begin{pmatrix} 0 & 0 \\ \text{pink} & 0 \end{pmatrix} & \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} \end{pmatrix}$$

**E**

Same pattern strength dynamics

**G**

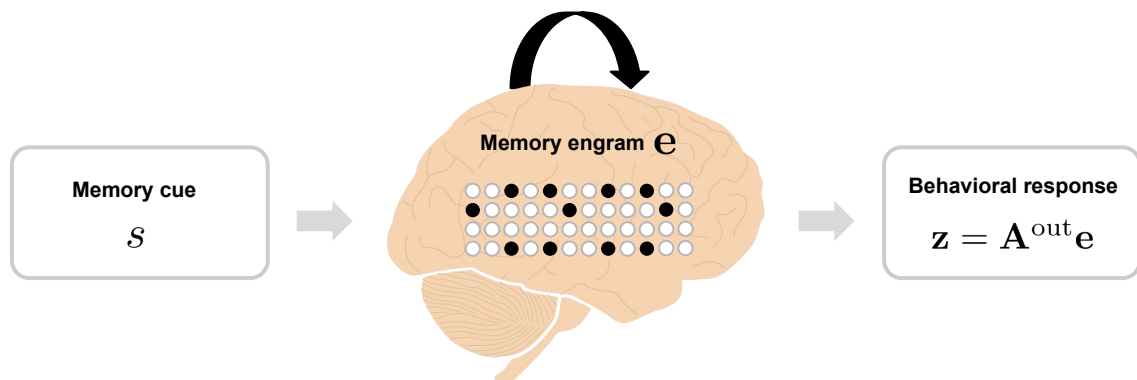
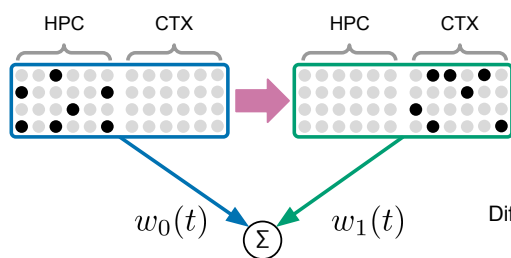
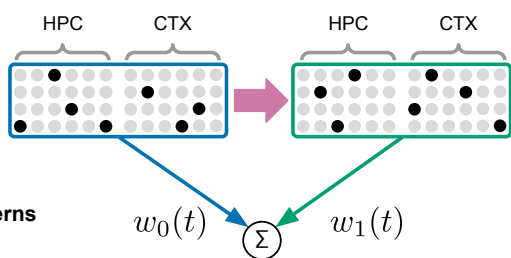
$$\mathbf{A} = \begin{pmatrix} \text{HPC} & \text{CTX} \\ \begin{pmatrix} \text{pink} & \text{pink} \\ \text{pink} & \text{pink} \end{pmatrix} & \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} \end{pmatrix}$$



A**Systems consolidation:**Reactivation changes
the engram

$$\frac{d}{dt} \mathbf{e} = -\tau^{-1} \mathbf{e} + \mathbf{A} \mathbf{e}$$

$\uparrow$
change = **forgetting** + **consolidation**

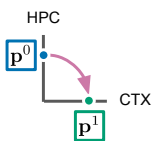
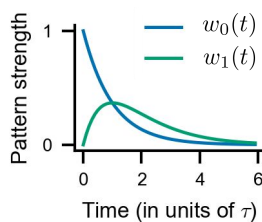
**B**Hippocampal pattern $\mathbf{p}^0$ Cortical pattern $\mathbf{p}^1$ **F**Distributed pattern $\mathbf{p}^0$ Distributed pattern $\mathbf{p}^1$ **C**

Different neural representation

$$\mathbf{e}(t) = \sum_{\alpha} w_{\alpha}(t) \mathbf{p}^{\alpha}$$

**D**

$$\mathbf{A} = \begin{pmatrix} \text{HPC} & \text{CTX} \\ \begin{pmatrix} 0 & 0 \\ \text{pink} & 0 \end{pmatrix} & \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} \end{pmatrix}$$

**E**Same
pattern strength
dynamics**G**

$$\mathbf{A} = \begin{pmatrix} \text{HPC} & \text{CTX} \\ \begin{pmatrix} \text{pink} & \text{pink} \\ \text{pink} & \text{pink} \end{pmatrix} & \begin{pmatrix} \text{HPC} \\ \text{CTX} \end{pmatrix} \end{pmatrix}$$

